METRIC

MIL-DTL-24674A(SH)
5 June 2018

SUPERSEDING
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21 August 1986

DETAIL SPECIFICATION

DISSOLVED OXYGEN KIT, NAVAL SHIPBOARD, METHOD C

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a naval shipboard, method C, water testing dissolved oxygen kit using an organic dye for the determination of dissolved oxygen in feedwater.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2. Government documents.

2.2.1 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

NAVAL SEA SYSTEMS COMMAND (NAVSEA) PUBLICATIONS

T9070-AL-DPC-020/077-2 - NAVSEA Hazardous Material Avoidance Process

(Copies of this document are available online via Technical Data Management Information System (TDMIS) at https://mercury.tdmis.navy.mil/ by searching for the document number without the suffix. Refer questions, inquiries, or problems to: DSN 296-0669, Commercial (805) 228-0669. This document is available for ordering (hard copy) via the Naval Logistics Library at https://nll.navsup.navy.mil. For questions regarding the NLL, contact the NLL Customer Service at nllhelpdesk@navy.mil, (866) 817-3130, or (215) 697-2626/DSN 442-2626.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN SOCIETY FOR QUALITY (ASQ)

ANSI/ASQ Z1.4 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of this document are available online at www.asq.org.)

Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 6630

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ASTM INTERNATIONAL

ASTM D5543 - Standard Test Method for Low-Level Dissolved Oxygen in Water

(Copies of this document are available online at www.astm.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Performance. The method C dissolved oxygen test kit (see 6.6) shall measure the amount of dissolved oxygen in deaerated feedwater that has a temperature of 21 to 38 °C (70 to 100 °F) between 0 and 40 parts per billion (p/b). The test results obtained shall be ±5 p/b. Feedwater contaminants or treatment chemicals, chlorides and hardness or phosphates, morpholine, Ethylenediaminetetraacetic acid (EDTA), hydrazine, and carbohydrazide shall not interfere with the measurement of dissolved oxygen. The kit shall be tested as specified in 4.3.2.

3.2 Physical characteristics.

3.2.1 Component parts. The method C dissolved oxygen test kit shall consist of the following:

a. One colorimetric comparator with eight liquid color standards representing 0, 5, 10, 15, 20, 25, 30, and 40 p/b dissolved oxygen, respectively.

b. One box of 30 ampoules.

c. One storage case.

3.2.1.1 Colorimetric comparator. Details and dimensions of the colorimetric comparator used for making the dissolved oxygen determinations shall be as shown on figure 1. The comparator shall consist of a transparent tube in which eight liquid-filled glass vial color standards are mounted and enclosed. There shall be a center space into which the test ampoule can be inserted for color comparison. The comparator shall have the color standards representing 0, 5, 10, 15, 20, 25, 30, and 40 p/b dissolved oxygen. These color standards shall range in color from clear to pink to reddish-pink. The comparator shall be tested as specified in 4.3.2.

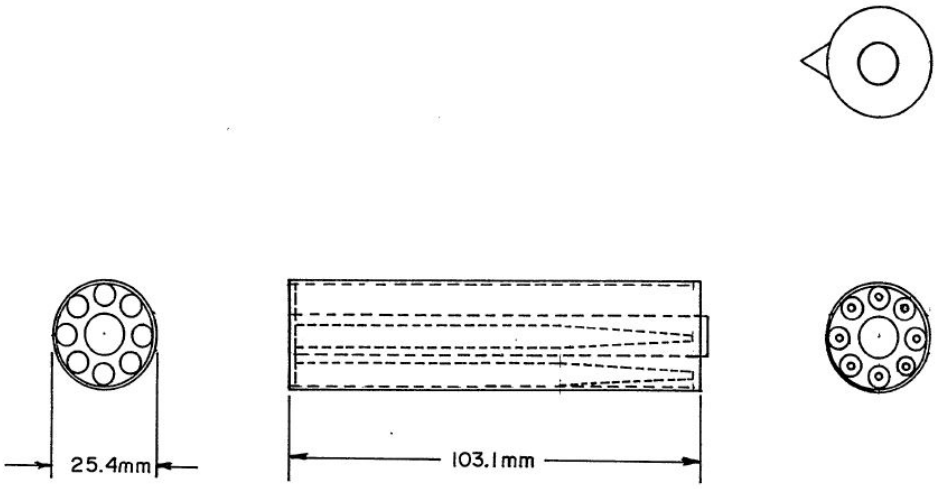


FIGURE 1. Comparator.

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3.2.1.2 Test ampoules. Details and dimensions of the test ampoules used in determining the dissolved oxygen content shall be as shown on figure 2. The test ampoules shall be made from glass, flat at one end and drawn to a tapered hermetically sealed closure at the other end. The tapered end shall be scored so that it breaks easily and evenly when placed in the snapper tool and broken. Each ampoule shall contain a color-forming reagent. This color-forming reagent shall be clear to pale yellow in color with no sediment present. If present, sediment should consist of a fine black powder that easily dissolves upon agitation of the ampoule. The ampoules shall contain a vacuum so they will self-fill when the tips are broken after they have been placed in the sample to be tested. When filled, the ampoules shall contain the sample and reagent. A small inert gas bubble shall be present in the broken ampoule to facilitate the mixing of the sample and reagent. The ampoules shall be packaged 30 to a box. The details and dimensions of the box shall be as shown on figures 2 and 3. The ampoules shall be tested as specified in 4.3.2.

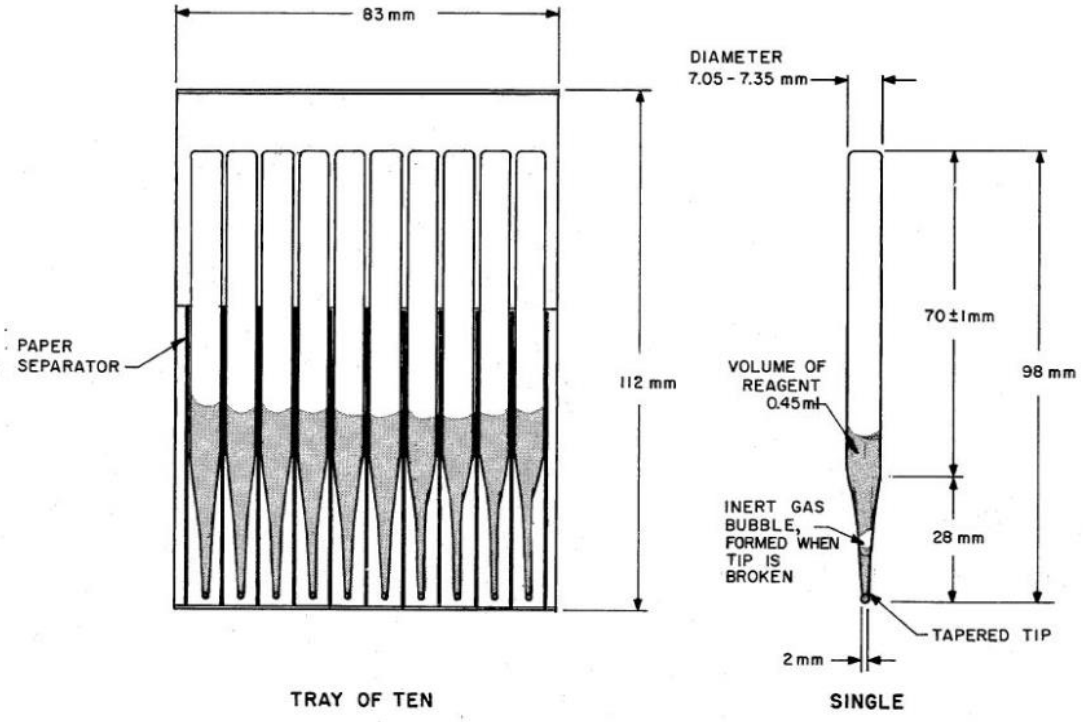


FIGURE 2. Glass ampoules.

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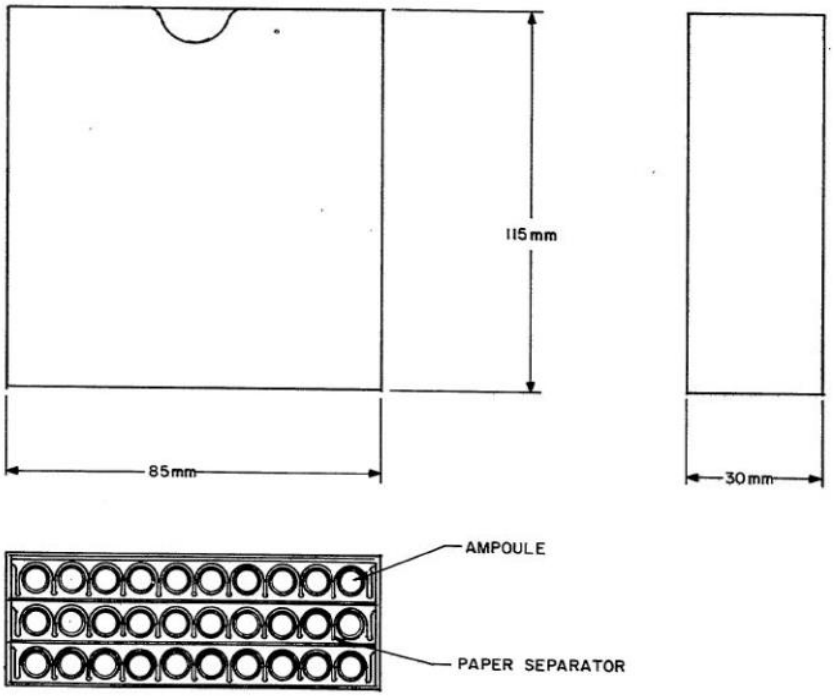


FIGURE 3. Box of ampoules.

3.2.1.3 Storage case. The case shall be one piece of molded plastic, with the upper and lower portion joined by a thin plastic hinge. The upper half shall contain a rectangular cavity into which a piece of foam or foam-like material may be inserted. This material shall have cutouts for storing a box of 30 ampoules, a snapper device (supplied separately), and one comparator. A snap-type locking device shall be on the upper and lower halves, with the hinged snap on the upper half and the catch on the bottom half.

3.3 Color standards. The color standards shall be fabricated to the dimensions shown on figure 1. The comparator shall have the date of manufacture stamped on it. The pink and reddish-pink colors shall yield the results specified in table I when tested in accordance with 4.3.2.

TABLE I. Dissolved oxygen solution values for colorimetric comparator.

Standard (p/b dissolved oxygen)	Accepted values (p/b)
0	0
5	5±5.0
10	10±5.0
15	15±5.0
20	20±5.0
25	25±5.0
30	30±5.0
40	35, 40

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3.4 Color standard stability. The colorant used in the manufacture of standards for this equipment shall exhibit in the completed standards a stability that shall give the specified values in table I for the duration of the comparator's shelf-life. The effective shelf-life of the colorimetric comparator shall be 2 years from the date of manufacture (see 4.3.1). The procedures for determining the accuracy of the comparator shall be as specified in 4.3.2.

3.5 Ampoule stability. Each box of ampoules shall have the month and year of manufacture stamped on it. The ampoules should have an effective shelf-life of 2 years from the date of manufacture (see 4.3.1). Procedures for determining the accuracy of the ampoule shall be as specified in 4.3.2.

3.6 Toxicity and prohibited materials.

3.6.1 Toxicity. When evaluated in accordance with 4.3, the kit shall pose no serious or high risk to the health of personnel or the environment when used for its intended purpose (see 4.2 and 6.3).

3.6.2 Prohibited materials. The kit shall not contain any chemicals categorized as "prohibited" in accordance with T9070-AL-DPC-020/077-2.

3.7 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

4. VERIFICATION

4.1 Conformance inspection. Conformance inspection shall be performed in accordance with inspection provisions set forth herein, as specified in table II. The requirements set forth in section 3 shall constitute minimum inspections to be performed by the supplier prior to Government acceptance or rejection. Unless otherwise specified (see 6.2), sampling shall be in accordance with ANSI/ASQ Z1.4. For purposes of sampling, an inspection lot for acceptance inspection and tests shall consist of all material submitted for inspection and delivery at one time. Failure of any test, by any sample, shall be cause for rejection of the lot represented.

TABLE II. Conformance inspection.

Inspection	Requirement	Verification
Performance	3.1	4.3.2
Visual examination	3.2.1.1 to 3.2.1.3 and 3.3	4.3.1 and 4.3.2
Comparator	3.4	4.3.1 and 4.3.2
Ampoule	3.5	4.3.1 and 4.3.2
Toxicity and prohibited materials	3.6	4.2

4.2 Toxicity and prohibited materials.

4.2.1 Toxicity. A Health Hazard Assessment (HHA) will be conducted to ensure conformance to 3.6.1 as specified (see 6.2). The Navy and Marine Corps Public Health Center (NMCPHC) will evaluate the kit using data provided by the manufacturer/distributor to the NMCPHC (see 3.6.1 and 6.3).

4.2.2 Prohibited materials. Prohibited materials shall be verified as specified (see 3.6.2 and 6.2).

4.3 Test methods and procedures.

4.3.1 Visual and dimensional examination. Each of the complete testing kits and components shall be visually and dimensionally examined to verify conformance with this specification. The colorimetric comparator and ampoules shall be visually examined to ensure there are no yellow or otherwise discolored chemicals. If any

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complete testing kit or component part in the sample contains one or more visual or dimensional defects, this shall be cause for rejection of the defective part or kit. If the number of defective testing kits or component parts in any sample exceeds the acceptance number for that sample (see 6.2), this shall be cause for rejection of the lot represented by the sample.

4.3.2 Comparator and ampoule performance. The kit shall be tested in accordance with ASTM D5543.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. This test kit is to be used for the determination of dissolved oxygen in boiler feedwater used on non-nuclear naval propulsion vessels. The kit is used with a snapper tool that fits onto sample coolers that comply with MIL-DTL-17557. Current stock of dissolved oxygen kits may be used until depleted; future requisitions for dissolved oxygen kits should refer to the current version of this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Requirements for an HHA (see 3.6.1 and 4.2.1).

c. Requirement for prohibited materials verification (see 3.6.2 and 4.2.2).

d. Sampling requirements if other than those specified in ANSI/ASQ Z1.4 (see 4.1 and 4.3.1).

e. Packaging requirements (see 5.1).

6.3 Toxicity evaluation. The NMCPHC requires sufficient information to permit an HHA of the product. Upon completion of the HHA, a copy will be provided by the NMCPHC to the Government for evaluation. The HHA process is described on the NMCPHC's website, http://www.med.navy.mil/sites/nmcphc/industrial-hygiene/Pages/health-hazard-assessment.aspx.

6.4 Shelf-life. This specification covers items where the assignment of a Federal shelf-life code is a consideration. Specific shelf-life requirements should be specified in the contract or purchase order, and should include, as a minimum, shelf-life code, shelf-life package markings in accordance with MIL-STD-129 or FED-STD-123, preparation of a materiel quality storage standard for type II (extendible) shelf-life items, and a minimum of 85 percent shelf-life remaining at time of receipt by the Government. These and other requirements, if necessary, are in DoDM 4140.27, DoD Shelf-life Management Program. The shelf-life codes are in the Federal Logistics Information System Total Item Record. Additive information for shelf-life management may be obtained from DoDM 4140.27, or the designated shelf-life Points of Contact (POC). The POC should be contacted in the following order: (1) the Inventory Control Points that manage the item and (2) the DoD Service and Agency administrators for the DoD Shelf-Life Program. Appropriate POCs for the DoD Shelf-Life Program can be contacted through the DoD Shelf-Life Management website: https://www.shelflife.dla.mil/.

6.5 Safety data sheets. Contracting officers will identify those activities requiring copies of completed Safety Data Sheets prepared in accordance with FED-STD-313. The pertinent Government mailing addresses for submission of data are listed in FED-STD-313.

6.6 Definition of dissolved oxygen test kit. The dissolved oxygen test kit measures the amount of dissolved oxygen present in deaerated feedwater at the p/b level. The kit consists of evacuated ampoules containing test chemicals and a standard colorimetric comparator.

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6.7 National stock numbers (NSNs). Table III provides a list of NSNs assigned that correspond to this specification. This list may not include all possible NSNs associated with this document.

TABLE III. List of NSNs.

Item Name	NSN	Shelf-Life Code (SLC)	Unit of Issue
Dissolved oxygen test kit (complete)	6635-01-044-6065	M	Kit
Ampoules, 0-40 p/b oxygen	6630-01-068-5210	M	PG (30)
Comparator, 0-40 p/b oxygen	6630-01-035-1864	M	EA

6.8 Subject term (key word) listing.

Ampoule

Color standard

Colorimetric comparator

Comparator

Deaerated feedwater

Organic dye

Reagent water

6.9 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Preparing activity:
Navy – SH
(Project 6630-2018-002)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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